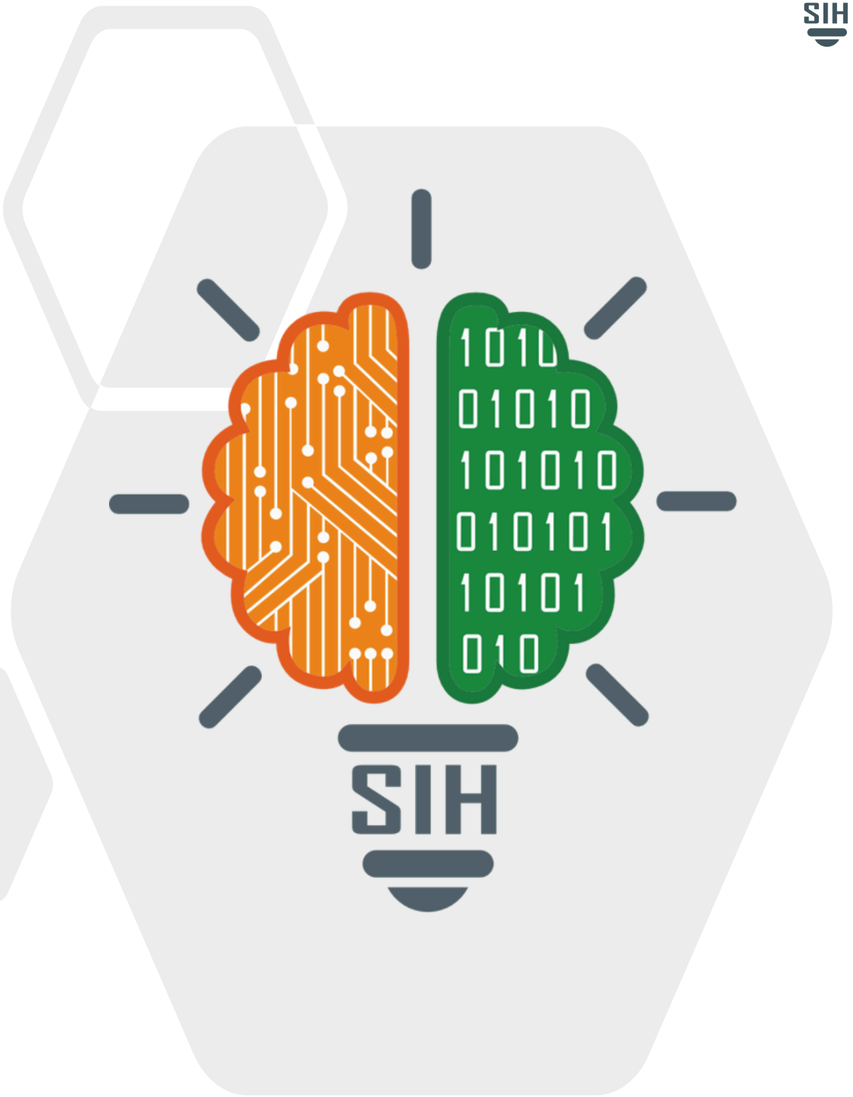
SMART INDIA HACKATHON 2026

SMART INDIA HACKATHON 2026

- **Problem Statement ID** –SIH26008
- **Problem Statement Title-**Belt Joint Rupture and Conveyor Belt Damages in Iron Ore Mining Industry: Intelligent Monitoring and Prediction of Conveyor Belt Joint Rupture and Damages in Iron Ore Mining Industry.
- **Theme-**Smart Automation
- **PS Category-** Hardware
- **Team Name -**Spark Spectrum



Spark Spectrum

BELTGUARD: AI-INTELLIGENT CONVEYOR BELT JOINT HEALTH MONITORING & PREDICTIVE PROTECTION SYSTEM

SMART INDIA HACKATHON 2026

PROBLEM

The joints on conveyor belts experience high loading conditions, vibrations, tensions, and extreme mining environments.

Small cracks, tears, and separations may occur slowly without being detected between manual checks, causing failure of the joints, spillage of materials, and downtime.



- Heavy Load
- Continuous vibration
- Belt Tension
- Abrasive Ore
- Dust & Harsh Condition

PROPOSED SOLUTION

BELTGUARD AI-Conveyor Joint Monitoring Using Edge-AI with the help of AI image inspection and multi-parameter sensing to assess the health of the joints and give early alerts about them.

- Image Analysis
- Vibration monitoring
- Temperature monitoring
- Tension monitoring
- Load monitoring

UNIQUENESS

AI-Based Vision: Detects belt-joint cracks, tears and wear automatically.

Multi-Sensor Fusion: Combines vision, vibration,sound,temperature, load and current data.

Health Score: Provides a real-time belt-joint health score ranges from 0-100.

Rupture Prediction: Predicts potential joint failure before catastrophic rupture.

Adaptive Control: Automatically adjusts conveyor speed through PLC + VFD.

Closed-Loop Protection: Sense → Predict → Control → Prevent.

HOW WE ADDRESS IT

JOINT INSPECTION
AI detects visible cracks, tears and joint separation.

SENSOR MONITORING
Monitors vibration, temperature, sound,tension and load.

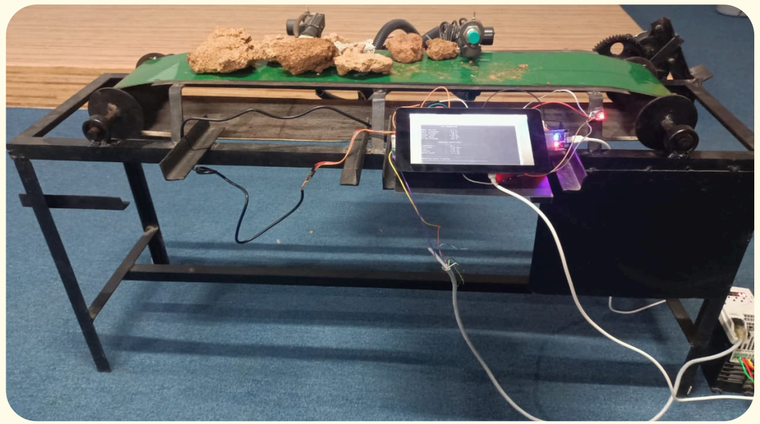
JOINT HEALTH INDEX
1.Normal
2.Warning
3.High Risk
4.Critical

EARLY ALERT
Notify before critical failure

PREVENTIVE ACTION
Schedule maintenance& reduce downtime

SAFER OPERATION
Relay based protection during critical conditions

OUR PROTOTYPE



Spark Spectrum

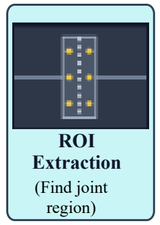
TECHNICAL APPROACH

SMART INDIA HACKATHON 2026

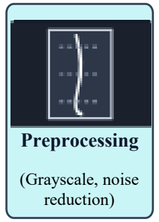
1 Data Collection (Sensors)

- Camera-Baslerace
- Sound-Ultratrack750
- Speed-VFD(CGDrive SK)
- Load -Uticell M340 Tension-Uticell M650
- Temperature-Thermal Scanner[Teledyne FLIR AX8]
- Vibration-A36D1OAC

2 Image Processing (OpenCV + AI)



ROI Extraction
(Find joint region)

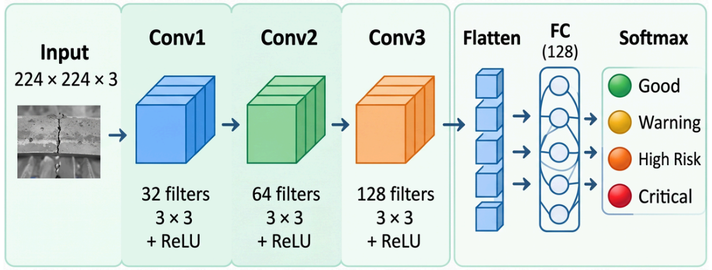


Preprocessing
(Grayscale, noise reduction)



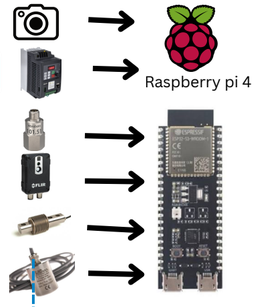
Damage Detection
• Crack • Tear
• Separation

3 CNN-Image Classification



4 Sensor Data Processing

- **Image Processing** (Filtering + Feature extraction)
- **Speed Analysis(VFD)** (Conveyor motor speed monitoring)
- **Vibration Analysis** (Filtering + Feature extraction)
- **Thermal Analysis** (Temperature mapping)
- **Load Analysis** (Load measurement + normalization)
- **Sound Analysis** (Detects abnormal acoustic patterns)



5 Data Fusion

Combine vision and sensor data for a complete view of the belt condition.

- **Vision** (damage area/type)
- **Vibration** (mechanical health)
- **Thermal** (heat condition)
- **Load** (measures material load)

Fusion Algorithm
(Weighted Analysis)

- **Tension** (detect excessice stress)
- **Speed** (operaing condition)
- **Sound** (acoustic condition)

6 Calculate Belt Health Score

Compute overall health score from fused data.

$$\text{Risk Score} = w_v(\text{CNN Risk}) + w_s(\text{Sensor ML Risk})$$

Vision Score (25%)	Vibration Score (10%)	Thermal Score (25%)	Load Score (25%)
Sound (10%)	Tension (5%)		

Belt Health Score (0 – 100)

7 Risk Calculation & Damage Classification

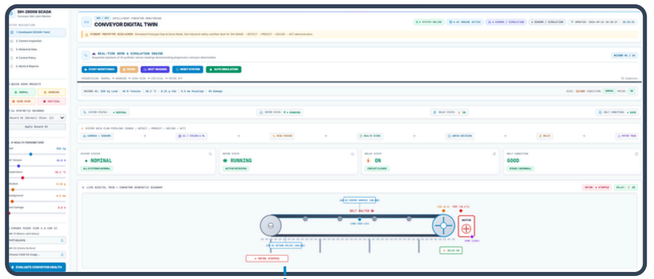
Score	Risk Level
90 – 100	GOOD
70 – 89	WARNING
40 – 69	HIGH RISK
0 – 39	CRITICAL

Convert health score to risk level and classify damage.

- **GOOD** (No significant damage)
- **WARNING** (Early deterioration)
- **HIGH RISK** (Significant damage)
- **CRITICAL** (Severe damage)

8 Dashboard (Monitoring Interface)

BELTGUARD



9 Alert to Admin

Send instant notification with detailed report.

- Risk level
- Damage details
- Sensor readings
- Recommended action

10 Admin Decision

Authorized admin reviews the dashboard and decides next action.

- Continue operation
- Schedule maintenance
- Stop conveyor

11 Relay Control

Raspberry Pi sends control signal to relay based on decision.



Raspberry pi 4
Premium Touch-Screen Case



12V channel Relay Module

12 Motor Control

The prototype uses a 12V DC motor
Motor ON / OFF based on decision.



12V DC motor

- on → Continue running
- off → Stop motor

13 CNN Mathematical Architecture & Feature Extraction

1. Convolutional Layer (Conv2D)

$$Z_{i,j,k} = \sum_m \sum_n \sum_c X_{i+m,j+n,c} \cdot W_{m,n,c,k} + b_k$$

2.Rectified Linear Unit (ReLU) Activation

$$A_{i,j,k} = \text{ReLU}(Z_{i,j,k}) = \max(0, Z_{i,j,k})$$

3. Softmax Classification Layer

$$P(y = c \mid \mathbf{z}) = \frac{e^{z_c}}{\sum_{j=1}^{C} e^{z_j}}$$

Spark Spectrum

FEASIBILITY AND VIABILITY

SMART INDIA HACKATHON 2026

Feasibility

Technical Feasibility
Raspberry Pi, sensors, camera, and relay can all be incorporated in the prototype.

AI & Computer Vision
OpenCV can analyze the belt joint and detect visible damage such as cracks, tears, and separation.

Real-Time Processing
The sensor data and joint images can be analyzed locally and used to derive a Belt Joint Health score.

Prototype Feasibility
Healthy and damaged joints can be used for testing the prototype.

Viability

Improvement in Safety
Early diagnosis reduces the possibility of an unexpected breakage of the belt joints.

Reduced Downtime
Maintenance can be done in advance of any serious rupture.

Low Cost for Maintenance
Diagnosing the joints early reduces the need for urgent repair work.

Operational Efficiency
Continuous monitoring gives real-time data regarding the belt status.

Potential Challenges & Risks

Lighting
Varying lighting conditions can affect image quality and detection accuracy.

Motion Blur & Belt Speed
High belt speed and movement can cause blurred images and missed detections.

Sensor Noise
Vibration, thermal and load sensors may produce noisy or inconsistent data.

False Damage Detection
AI may sometimes misclassify healthy joints as damaged or miss real damage.

Strategies to Overcome

Controlled LED Lighting & Image Preprocessing
Use proper lighting, camera enclosure and image enhancement (denoising, contrast adjustment).

Fixed Camera & Synchronized Capture
Use a stable camera mount and synchronize image capture with belt movement (appropriate shutter speed).

Sensor Calibration & Data Filtering
Calibrate sensors regularly and apply filtering/averaging to remove noise and improve accuracy.

Healthy/Damaged Dataset & AI Tuning
Train and test the model with diverse samples (healthy, damaged, dusty, different joint conditions) to improve reliability.

BELTGUARD
SMART BELT • SAFER MINING

VIDEO LINK

Spark Spectrum

IMPACT AND BENEFITS

SMART INDIA HACKATHON 2026

BENEFITS OF THE SOLUTION

Safer Inspection
Fewer manual inspections required on conveyor system.

Planned Maintenance
Solve problems before they arise.

Less Downtime
Operate the conveyor for more time.

Lower Emergency Cost
Prevent expensive malfunctions and maintenance.

Reduce Material Waste
Increase belt lifespan, avoid replacements.

IMPACTS OF THE SOLUTION

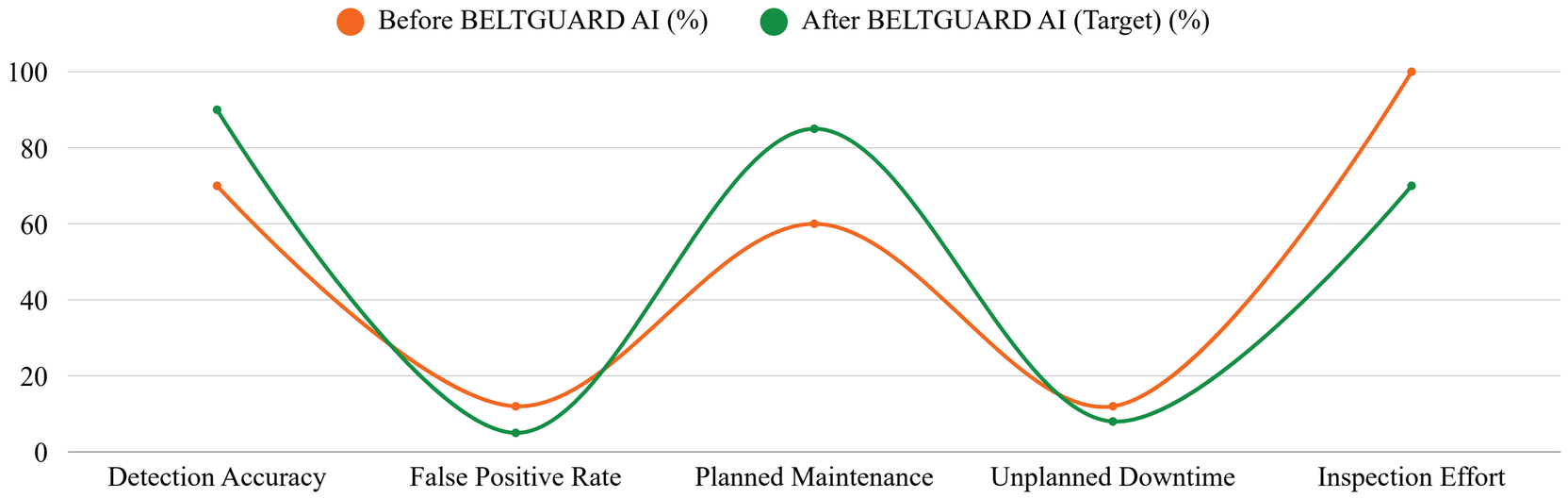
>=90%
Detection Accuracy

<=5%
False Positive Rate

>=25 min
Warning Lead Time

10-30%
Reduction in unplanned downtime

>=30%
Less Inspection Effort



Spark Spectrum

RESEARCH AND REFERENCES

SMART INDIA HACKATHON 2026

1. Field Visit / Industry Observation

- Conveyor belt operation in a mining environment.
- Studied belt joint/splice area and working conditions
- Heavy material, dust, vibration and continuous operation effect observed
- Explored current inspection/monitoring of joint condition
- Realized the importance of early detection of joint deterioration



2. Literature Survey

Field visit and Reviewed research concerning:

We conducted an industrial field visit to **Thriveni Earthmovers Private Limited in Salem,Tamil Nadu** to understand real-time conveyor operations, maintenance practices, and the practical challenges faced by the industry. Discussions with industry personnel helped us identify belt-joint damage, inspection difficulties, downtime, and maintenance challenges, which guided us in developing our proposed intelligent monitoring and predictive maintenance system.

Zheng et al. (2024)
Novel mining conveyor monitoring system based on quasi-distributed optical fiber accelerometer array and self-supervised learning
Mechanical Systems and Signal Processing, 221, 111697.
Link

Zhang et al. (2021)
Deep learning-based damage detection of mining conveyor belt
Measurement, 175, 109130.
Link

Li et al. (2021)
Health State Prediction and Performance Evaluation of Belt Conveyor Based on Dynamic Bayesian Network in Underground Mining
Shock and Vibration, 2021, Article 6699611.
Link

Młynarek et al. (2020)
A Diagnostics of Conveyor Belt Splices
Applied Sciences, 10(18), 6259.
Link

Guo et al. (2023)
Machine vision based damage detection for conveyor belt safety using Fusion knowledge distillation
Alexandria Engineering Journal, 71, 161–172.
Link

3. Identified research gap

Current research focuses on individual aspects like visual inspection, splice diagnostics or sensor monitoring. In our project we study the integration of these approaches for ongoing evaluation of conveyor joint health.